Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/20/2024
PI Name:	Basner, Mathias M.D., Ph.D.		
Project Title:	Long-Term Brain Structural and Functional Consequences of Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H) (2) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS) (3) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	basner@pennmedicine.upenn.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	215-573-5866
Organization Name:	University of Pennsylvania		
PI Address 1:	Department of Psychiatry, Division of Sleep and Chronobiology		
PI Address 2:	423 Service Dr, 1013 Blockley Hall		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	03/15/2021	**End Date:**	09/30/2033
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2033 per NSSC documentation (Ed., 6/6/23). NOTE: End date changed to 03/31/2029 per L. Juliette/JSC (Ed., 5/3/22).		
Key Personnel Changes/Previous PI:	David F. Dinges retired at the end of June 2024 and was removed from the proposal. In his stead, Christopher W. Jones, PhD was added. He is a Research Assistant Professor with expertise in astronaut behavioral health.		

COI Name (Institution):	Gunga, Hanns-Christian M.D. (Charite - Universitatsmedizin Berlin, Germany) Gur, Ruben Ph.D. (The Trustees of the University of Pennsylvania) Hartley, Tom Ph.D. (University of York, United Kingdom) Kuehn, Simone Ph.D. (Max Planck Institute for Human Development, Berlin, Germany) Riecke, Bernhard Ph.D. (Simon Fraser University, Canada) Roalf, David Ph.D. (University of Pennsylvania) Bell, Suzanne Ph.D. (NASA Johnson Space Center) Stangl, Matthias Ph.D. (University of California, Los Angeles) Wolbers, Thomas Ph.D. (German Center for Neurodegenerative Diseases, Germany) Stahn, Alexander Ph.D. (Charite - Universitatsmedizin Berlin, Germany) Christopher, Jones Ph.D. (University of Pennsylvania)
Grant/Contract No.:	80NSSC21K1698
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Ocular and brain structural changes present during and after ~6-month standard duration missions to the International Space Station (ISS). Approximately 70% of crew develop optic disc edema, chorioretinal folds, globe flattening, and/or a hyperopic shift in refractive error, which has been termed spaceflight associated neuro-ocular syndrome (SANS). Brain structural changes include ventricular enlargement, regional shifts in free water, and an upward shift of the brain. The long-term consequences of spaceflight on the brain and eye—either related to or independent of SANS—are largely unknown. Completion of the following specific aims will generate objective data that describe long-term consequences of spaceflight on the structure and function of the eye and brain. Specific Aim 1: Characterize the recovery and long-term ocular structural changes up to 5 years following 6-month spaceflight missions. Specific Aim 2: Characterize the recovery and long-term ocular functional changes up to 5 years following 6-month spaceflight missions. Specific Aim 3: Characterize the recovery and long-term brain structural changes up to 5 years following 6-month spaceflight missions. Specific Aim 4: Characterize the recovery and long-term brain functional changes up to 5 years following 6-month spaceflight missions. Specific Aim 5: Characterize the structural and functional changes to the eye, brain, and cognition in crewmembers who participated in past missions, calling them back several years after their flights. Specific Aim 6: Integration and characterization of the relationships across ocular and brain measures, crewmembers, and time. This project will evaluate the persistence of spaceflight-induced ocular, brain, and behavioral changes, as well as quantify their association with long-term health and performance. We will quantify regional changes and recovery in brain structure, brain fluid distribution, changes in brain activation and functional connectivity, glymphatic clearance, and optic nerve head structure; we will also assess visual function, sensorimotor function, cognitive function, and quality of life. Finally, we will determine the molecular signatures of brain changes. Investigating these parameters is critically important as we prepare to extend the duration and distance of space travel. This project will be comprised of two arms: (1) a prospective spaceflight arm that includes a 5-year postflight follow-up and (2) a call-back arm in which crew who have flown a ~6-month standard duration mission (or longer) > 5 years earlier will be invited to participate. In the prospective flight arm, long-term structural and functional changes to the eye and brain will be characterized for up to 5 years in 10 crewmembers completing approximately 6-month spaceflight missions. Crewmember data will be compared to similar data from 10 age- and sex-matched controls. Eye and brain structure and function will be assessed before, during, and after spaceflight. The call-back arm will recruit 20 crewmembers who have previously participated in a long-duration (≥ 6 month) mission and 20 controls with no spaceflight experience to further characterize the lasting structural and functional effects of long-duration spaceflight in a cross-sectional population. Ocular testing will include optical coherence tomography, electroretinography, intraocular pressure, visual field analysis, optical biometry, and orbital magnetic resonance imaging (MRI). Multi-modal brain MRI will assess changes in brain activation during tasks, functional connectivity via task and resting state functional MRI, brain volume, lateral ventricle size, and pituitary gland deformation. Comprehensive neurological examination and neuropsychology testing will be performed to correlate MRI findings with cognitive and neurological functioning. Blood samples will be obtained to identify key neurotrophins and neurodegenerative biomarkers. This project has been combined with the work of three other Principal Investigators (PIs) who responded to the same solicitation with independent proposals. The protocols and aims from all four PIs have been integrated into a revised proposal that was delivered to NASA Human Health Countermeasures (HHC) and Human Factors and Behavioral Performance (HFBP) Element Scientists in January 2022. For information on the four related investigations, see: • The Long-Term Consequences of Spaceflight on Brain and Eye Health (PI: Bershad). • Long-Term Brain Structural and Functional Consequences of Spaceflight (PI: Basner, this taskbook entry). • Investigating Long-term Structural and Functional Changes in the Eye and Brain After Spaceflight (PI: Macias). • Recovery Timeline of Spaceflight-Induced Central Nervous System Changes (PI: Seidler).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Detailed astronaut follow-up after return from spaceflight has historically been limited to a few weeks, which prevents conclusions about long-term health consequences of astronauts, especially after longer stays in space. This study will follow astronauts for up to 5 years after >=6 months missions. The research partially translates to similar stressful long-term exposure situations on Earth.

	This “Long-Term Health” (LTH) Virtual NASA Specialized Center of Research (VNSCOR) has a ground and flight component, and therefore two Research Operations and Integration (ROI) coordination teams. We participated in bi-weekly meetings with the LTH Co-PI teams to coordinate the overall implementation effort of the ground and flight components of the project. The “LTH Call Back” retrospective arm will focus on investigating the structural and functional changes in the eye, brain, and cognition in astronauts who previously participated in long-duration missions, calling them back at least 5 years after their previous flights to determine changes that may endure long after return from spaceflight. We have limited involvement in this arm of the study. Major accomplishments during this reporting period for this sub-study include: • The Lifetime Surveillance of Astronaut Health (LSAH) data share request was approved by the LSAH Advisory Board. • An LSAH epidemiologist was added to the study team to help with data analysis. • The MRI facility contract was finalized, and two MRI dry runs were completed at Houston Physicians’ Hospital. • One internal dry run was completed in the Cardiovascular and Vision Laboratory. • The neuropsychological and neurological Co-PI team completed a dry run at NASA Johnson Space Center (JSC). • The Crew Office approved the Informed Consent Briefing (ICB). • Recruitment efforts were initiated by ROI and are ongoing. • The Test Readiness Review (TRR) package was developed and is undergoing internal review.
Task Progress:	Ongoing forward work with ROI for “LTH Call Back” includes: • Approval of the TRR package (planned for December 2024). • Initiate recruitment and testing of astronauts and matched control TSS participants. The “Brain Health” component is prospective and will investigate the development and recovery of structural and functional changes in the eye, brain, and cognition in astronauts participating in standard or extended duration missions and will follow them out 5 years postflight. Major accomplishments for the Flight Study during this reporting period include: • ROI submitted the Feasibility Assessment to the HHC and HFBP Elements and the sub-study was approved for Select for Flight in December 2024. • The LSAH data share request was approved by the LSAH Advisory Board. • An LSAH epidemiologist was added to the study team to help with data analysis. Ongoing future work with ROI for “Brain Health” includes (pending select for flight determination by HRP): • Completion of two MRI dry runs for the Brain Health MRI scanning protocol. • Development and approval of a TRR package (targeting January 2025). • Development and approval of ICB presentation. • Pitch ICB starting with Crew-12.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Abstracts for Journals and Proceedings	Carter KJ, Basner M, Bershad EM, Seidler R, Stahn AC, Laurie SS, Macias BR. "Investigating long-term structural and functional changes to the eye and brain after spaceflight." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024